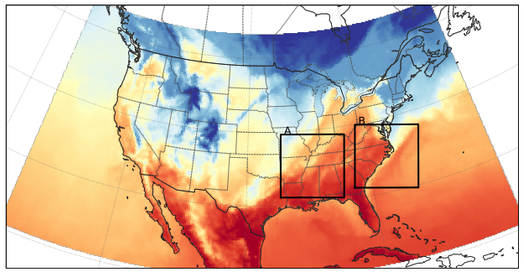
MERRA21C-ML: GENERATIVE CONUS DOWNSCALING

GEOS-FP (~25 km) → Conditional flow matching → GEOS grid (~3 km)

A COARSE INPUTS

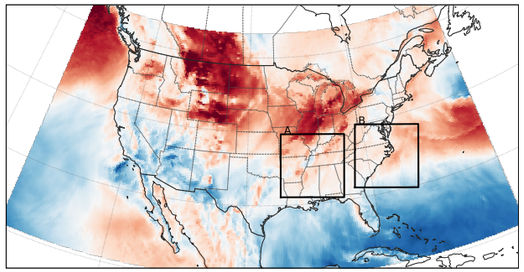
GEOS-FP · ~25 km

Temperature (°C)



0 15 30

Zonal wind (m s⁻¹)



-5 0 5

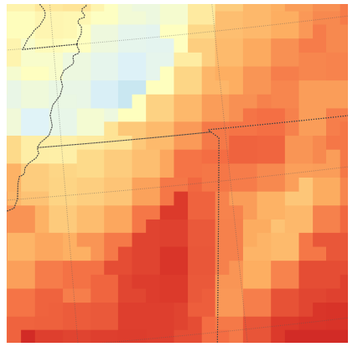
Weather at t − 1 h and t

Terrain · location · time

B RESOLUTION

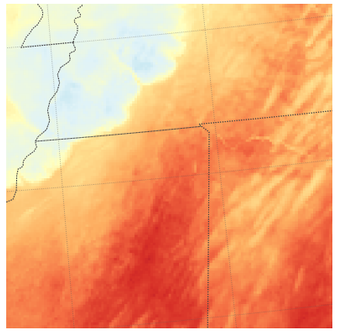
Same extent · same scale

GEOS-FP · ~25 km



10 17.5 25

GEOS replay · ~3 km



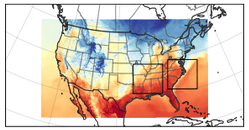
10 17.5 25

Temperature (°C) · 3× zoom

C GENERATED FIELDS

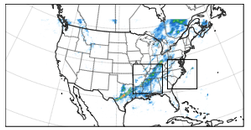
One ensemble member · ~3 km

T₂m (°C)



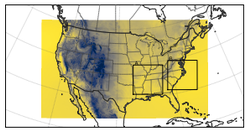
0 15 30

Rain (mm h⁻¹)



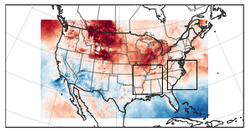
0.1 4 64

pₛ (hPa)



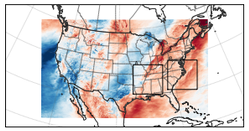
800 1000

u₁₀ (m s⁻¹)



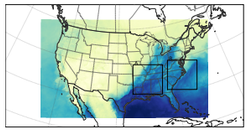
-5 0 5

v₁₀ (m s⁻¹)



-5 0 5

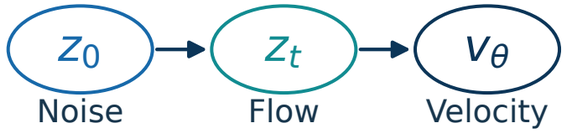
q₂m (g kg⁻¹)



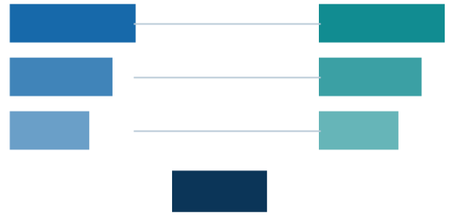
2 8 14

D FLOW MATCHING

Frozen regression conditioner



Conditional U-Net



Weighted velocity loss



Core + halo

Area + channels

Ensemble sampling



Independent noise seeds